Literature-Implied Negative Answer to the Average-Distortion Extension

arXiv:2310.01868 target; arXiv:2410.21931 support

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Status. *Literature-implied answer, negative for the dimension-refined extension in Discussion item 3 of arXiv:2310.01868.* The relation is an agent-identified implication of the proposition labelled `prop:Rn more genral` in arXiv:2410.21931; the supporting authors do not appear to explicitly cite or answer this discussion item.

Source question. In Section "Discussion and open problems," item 3, Eskenazis asks whether the bound of Theorem labelled `thm:main` can be extended from bi-Lipschitz embeddings to average-distortion embeddings. The theorem gives, for finite-dimensional normed X,

$$\mathsf{c}_X(\{-1,1\}^n) \gtrsim \sup_{1\le p\le 2} \frac{n}{\mathsf{T}_p(X)\min\{n,\dim(X)\}^{1/p}}.$$

The same discussion notes that IVV-type arguments imply only the dimension-free average-distortion lower bound

$$\mathsf{T}_p(X)^{-1}n^{1-1/p}.$$

Supporting theorem. Chang–Naor–Ren, arXiv:2410.21931, Proposition labelled `prop:Rn more genral`, states that for $1 \le q \le 2$, the smallest $D \ge 1$ for which ℓ_q^n embeds into $\mathbb{R}$ with p-average distortion D is, up to universal constants,

$$\sqrt{\max\{1, n/p\}}.$$

Their definition of p-average distortion is the standard Rabinovich–Naor one: for every Borel probability measure on the source, there is a Lipschitz map satisfying the required L_p-average lower separation after scaling.

Implication for the source question. Take $X = \mathbb{R}$. Then $\dim X = 1$ and $\mathsf{T}_2(\mathbb{R}) = 1$. The $p = 2$ term in the source theorem bound is therefore of order

$$\frac{n}{\mathsf{T}_2(\mathbb{R})\min\{n,1\}^{1/2}} = n.$$

If that dimension-refined lower bound extended to quadratic average distortion, then $\{-1,1\}^n$ would require quadratic average distortion $\gtrsim n$ into $\mathbb{R}$.

But the supporting proposition with $q = 1$ and $p = 2$ gives an embedding of ℓ_1^n into $\mathbb{R}$ with quadratic average distortion $O(\sqrt{n})$. Restricting that embedding to the subset $\{-1,1\}^n$ preserves the same average-distortion upper bound for every probability measure on $\{-1,1\}^n$, by extending the measure by zero to ℓ_1^n. On $\{-1,1\}^n$, the inherited ℓ_1 metric is exactly twice the Hamming metric, and scaling the source metric changes neither the order nor the validity of an average-distortion estimate.

Consequently,

$$\mathrm{Av}_2(\{-1,1\}^n,\mathbb{R}) \lesssim \sqrt{n},$$

whereas the proposed dimension-refined extension would force

$$\mathrm{Av}_2(\{-1,1\}^n,\mathbb{R}) \gtrsim n.$$

For large n, these are incompatible. Thus the source theorem's dimension-refined lower bound does not extend to average-distortion embeddings in this form.

Scope and limitations. This negative answer targets the low-dimensional improvement in Theorem labelled `thm:main`. It does not refute the IVV-style average lower bound quoted in the source discussion: for $X=\mathbb{R}$ and $p=2$, that bound is order $\sqrt{n}$, which is consistent with the supporting proposition.

References

[1] Alexandros Eskenazis. *Some geometric applications of the discrete heat flow*. arXiv:2310.01868, 2023.

[2] Alan Chang, Assaf Naor, and Kevin Ren. *Random zero sets with local growth guarantees*. arXiv:2410.21931, 2024.